

Bilal Ahmad Kalas

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Profile

M.Sc. Mathematics graduate with broad interests in pure mathematics, particularly algebra, topology, and number theory. My master's dissertation focused on finite transformation semi-groups, providing a strong foundation in abstract algebra and mathematical proof techniques. I am seeking doctoral research in pure mathematics, with particular interests in algebra, commutative algebra, algebraic geometry, arithmetic geometry, representation theory, and their interactions with topology and number theory.

Education

Integrated B.Sc.-M.Sc. Mathematics
Central University of Kashmir, India

2019–2024

Master's Dissertation

Dissertation: Ranks and Presentation for Finite Transformation Semigroups

[PDF](#)

Supervisor: Dr Aftab Hussain Shah (Associate Professor)

- Studied rank and presentation problems for finite order-preserving transformation semi-groups.
- Based on the work of Joerg Koppitz and Somuek Worawiset on transformations with one fixed point.
- Focused on semigroup presentations, generating set, ranks, and structural & combinatorial properties of finite transformation semigroups.
- JavaScript and python implementation for computing cardinality and rank of order-preserving finite transformation semigroups with one fixed point.
- JavaScript implementation for deriving presentation of order-preserving finite transformation semigroups with '1' as the only fixed point.
- Strengthened my ability to independently study research literature, develop rigorous mathematical arguments, and communicate advanced mathematical ideas through regular research presentations.

Research Interests

Algebra, Commutative Algebra, Algebraic Geometry, Arithmetic Geometry, Number Theory, Topology, Representation Theory, Galois Theory, Semigroup Theory

Teaching Experience

Private Mathematics Tutor

2024-2025

Skills

Relevant Coursework

Abstract Algebra, Commutative Algebra, Field and Galois Theory, Topology, Algebraic Topology, Number Theory, Linear Algebra, Differential Geometry, Measure Theory, Semigroup Theory, Functional Analysis, Graph Theory

Academic Skills

Mathematical proof writing, mathematical exposition, independent study, reading and analysis of mathematical literature.

Technical skills

Latex, Python, HTML/CSS, JavaScript, C

Languages

- Hindi/Urdu – Native
- Gojri – Native
- Kashmiri – Native
- English – Fluent

References

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